

COURSE CHEMV51: FOOD CHEMISTRY & ADULTERATION

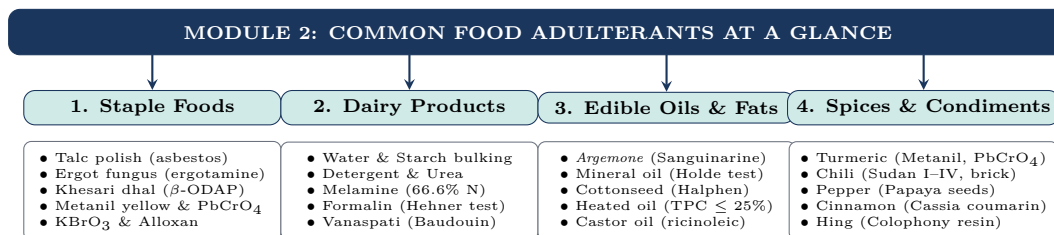
Module 2: Common Adulterants in Food

Concise Exam Notes: Mechanisms, Reactions, Tests, Mnemonics & Solved Questions

1 Overview of Food Adulteration

Definition: Fraudulent addition of substandard/toxic substances or removal of vital nutritional constituents from food commodities to maximize illicit profit or conceal spoilage.

- **Intentional (EMA):** Deliberate commercial fraud (watering milk, urea, Vanaspati in ghee, synthetic colorants).
- **Incidental:** Unintentional contamination (ergot mold sclerotia, pesticide residues, heavy metal leaching).



2 Adulteration in Staple Foods (Grains, Pulses, Flours)

2.1 Grains (Wheat, Rice, Bajra)

- **Talc / Soapstone** ($\text{Mg}_3\text{Si}_4\text{O}_{10}(\text{OH})_2$): Imparts white shine to rice/pulses. Stomach irritant; often carries tremolite asbestos fibers (IARC Group 1). *Detection:* High acid-insoluble ash ($> 0.5\%$).
- **Mineral Oil Glaze:** Non-saponifiable hydrocarbons prevent moisture loss; blocks absorption of fat-soluble vitamins (A, D, E, K). *Detection:* Bright blue-violet fluorescence under UV (365 nm).
- **Ergot Sclerotia (*Claviceps purpurea*):** Purplish-black fungal bodies contaminating rye, wheat, bajra. Contain peptide alkaloids (**Ergotamine**, **Ergocristine**) that cause intense peripheral vasoconstriction $\rightarrow$ **Ergotism** / **St. Anthony's Fire** (dry gangrene of digits, severe burning, uterine contractions/abortions).

Salt Floatation Test for Ergot

Suspend grains in a 20% NaCl **brine solution** (sp. gr. ≈ 1.15). Clean sound grains sink; light spongy ergot sclerotia float on the surface and are skimmed off.

- **Datura Seeds:** Kidney-shaped black seeds with tropane alkaloids (**Hyoscyamine**, **Scopolamine**). Anticholinergic toxidrome (mydriasis, extreme tachycardia, dry mouth, delirium).

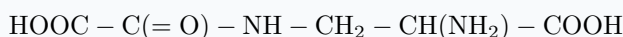
2.2 Pulses (Legumes)

Khesari Dhal (**Lathyrus sativus**) and Neurolathyrism

Fraud: Blended into expensive Arhar/Toor dhal (**Cajanus cajan**) and Chana dhal.

Morphology: Wedge-shaped, trapezoidal, slant-cut seeds vs. round Arhar dhal.

Neurotoxin: β -N-oxalyl- α , β -diaminopropionic acid (**β -ODAP** / **BOAA**).



Mechanism: Structural analogue of glutamate; hyper-activates post-synaptic ****AMPA receptors**** on Betz cells and spinal motor neurons $\rightarrow$ massive Ca^{2+} influx $\rightarrow$ oxidative motor neuron lysis $\rightarrow$ **Neurolathyrism** (irreversible spastic paraplegia of legs, scissoring gait).

Detoxification: β -ODAP is water-soluble; boiling seeds in water (1 : 10) for 60 min and draining removes $> 85\%$ toxin.

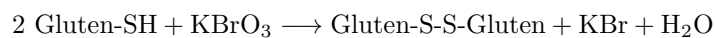
- **Metanil Yellow:** Synthetic monoazo dye added to color inferior pulses and turmeric. Cleaved by gut azoreductases into toxic **p-aminodiphenylamine** (causes testicular seminiferous tubule atrophy).

Extract with water+conc. HCl $\rightarrow$ **Persistent Magenta / Pink-Red Color** ($\lambda_{\text{max}} \approx 530 \text{ nm}$, quinoid dication)

- **Lead Chromate** (PbCrO_4): Industrial paint pigment on turmeric and pulses. Pb^{2+} inhibits δ -ALAD in heme synthesis $\rightarrow$ microcytic anemia, saturnism, and encephalopathy. *Detection:* Ash sample + HNO_3 + KI $\rightarrow$ **golden-yellow** $\text{PbI}_2 \downarrow$.

2.3 Flours (Atta, Maida)

- **Potassium Bromate** (KBrO_3): Strong oxidizing agent. Converts gluten thiol ($-\text{SH}$) to disulfide bridges ($-\text{S}-\text{S}-$):



Inflates bread loaf volume. IARC 2B carcinogen (renal cell tumors); banned by FSSAI. Spotting flour with 0.5% KI + 2M HCl gives **black-purple spots** (I_2 + starch).

- **Alloxan** ($\text{C}_4\text{H}_2\text{N}_2\text{O}_4$): Bleaching byproduct in maida. Enters pancreatic β -cells via **GLUT2** $\rightarrow$ reduced to dialuric acid $\rightarrow$ generates cytotoxic hydroxyl radicals ($\cdot\text{OH}$) via Fenton reaction $\rightarrow$ necrotic β -cell death $\rightarrow$ **Type-1 Diabetes**.

3 Adulteration in Dairy Products

3.1 Synthetic Milk Formulation & Rapid Screening Tests

Synthetic milk is prepared using vegetable oil (fat phase), detergent/LAS (emulsifier/foam), urea (specific gravity/NPN), neutralizers (NaOH, NaHCO₃ to prevent boiling curdling), starch (SNF/viscosity), and formalin (preservative).

Adulterant	Rapid Chemical Test & Reagents	Diagnostic Positive Observation & Note
Water Dilution	Lactometer Reading / Cryoscopy	Lactometer < 28 at 20°C; freezing point shifted toward 0°C (norm: -0.540°C).
Starch / Dextrin	1% I ₂ /KI Iodine Reagent	**Intense deep blue-black color** (polyiodide-amylose complex).
Urea	DMAB reagent (<i>p</i> -dimethylaminobenzaldehyde)	**Intense bright yellow-green color** (pure milk gives faint yellow).
Detergents	Methylene Blue dye + Chloroform shake	**Lower chloroform layer turns blue** (anionic surfactant complex).
Neutralizers	0.1% Rosalic Acid in ethanol	**Rose-red color** (pure milk gives pale brownish-orange).
Formalin (37%)	**Hehner's Test** (conc. H ₂ SO ₄ + trace Fe ³⁺)	**Violet / purple ring at interface** (condensation with tryptophan in casein).

Melamine Triazine Fraud (C₃H₆N₆)

Analytical Flaw: Melamine has ****66.6% nitrogen by mass****. Standard Kjeldahl testing calculates crude protein as %N × 6.38, reading melamine as protein (1 g melamine in 1 L watered milk inflates protein by +4.25%).

Toxicity: Melamine pairs via hydrogen bonding with cyanuric acid to form insoluble ****[Melamine · Cyanuric Acid]**** co-crystalline spherulites ($K_{sp} \approx 10^{-9} \text{ M}^2$) in renal tubules → ****Obstructive urolithiasis and acute kidney injury****.



Figure: Hehner's Test Tube for Formalin

3.2 Ghee and Butter Adulteration

- **Vanaspati (Hydrogenated Oil) in Ghee:** Catalytic partial hydrogenation (H₂/Ni) converts cis-unsaturated fats to saturated and *trans*-fatty acids (**elaidic acid**). Lowers Reichert-Meissl (RM) value from normal 26 – 32 to < 1.0.
- **Baudouin Test:** Vanaspati legally contains 5% sesame oil. Conc. HCl hydrolyzes sesame ****sesamolin**** to ****sesamol****, which condenses with added ****furfural**** → ****crimson-rose red color**** in acid layer. Pure ghee yields no color.

4 Adulteration in Edible Oils and Fats

Argemone mexicana Poisoning (Epidemic Dropsy)

Seeds: Smooth round mustard seeds vs. rough, pitted *Argemone* seeds with a ****pointed apical beak****.

Toxic Alkaloid: ****Sanguinarine**** and Dihydrosanguinarine.

Pathophysiology: (1) Sanguinarine inhibits membrane ****Na⁺/K⁺-ATPase**** → cellular Na⁺ retention and edema; (2) Hyper-activates ****eNOS****, releasing massive ****Nitric Oxide (NO)**** → systemic capillary hyper-permeability and fluid extravasation.

Clinical Signs (Dropsy): Bilateral lower extremity pitting edema with erythema, secondary open-angle glaucoma (trabecular capillary dilation), and high-output congestive heart failure.

Detection: (a) Oil + conc. HNO₃ → ****crimson-red ring****; (b) Saponified oil + HCl + FeCl₃ → reddish-brown needle crystals; (c) TLC under UV (366 nm) → bright ****golden-yellow/orange fluorescence**** ($R_f \approx 0.42$).

- **Mineral Oil (Holde's Saponification Test):** Vegetable oils saponify with alcoholic KOH to water-soluble soaps. Petroleum hydrocarbons cannot saponify; adding boiling water causes ****persistent milky turbidity****.
- **Cottonseed Oil (Halphen Test):** Contains toxic ****gossypol**** and cyclopropenoid fatty acids (malvalic, sterculic). Heated with 1% sulfur in CS₂ at 110°C produces a persistent ****deep rose-red color****.

- **Used Cooking Oil (UCO):** Deep frying ($> 180^{\circ}\text{C}$) forms toxic aldehydes (acrolein, 4-HNE) and polymers. **FSSAI statutory threshold: Total Polar Compounds (TPC) $\leq 25\%$.** Oil exceeding 25% TPC is banned.

5 Adulteration in Spices and Condiments

Spice Matrix	Common Adulterants	Diagnostic Screening Test	Toxicology / Hazard Impact
Turmeric Powder	Metanil Yellow, Lead Chromate	Conc. $\text{HCl} \rightarrow$ magenta-red; $\text{KI} \rightarrow$ yellow $\text{PbI}_2 \downarrow$	Testicular damage, saturnism, anemia.
Chili Powder	Sudan I–IV dyes, brick dust, red lead	Brick settles in water; Sudan detected by TLC	Sudan dyes are genotoxic IARC carcinogens.
Black Pepper	Dried Papaya Seeds	Floatation: Papaya seeds float; pepper sinks	Ingestion of toxic carpine alkaloid.
Cinnamon	Cassia Bark (*C. cassia*)	Cassia turns deep blue with iodine decoction	Cassia has Coumarin (0.2–1%) $\rightarrow$ liver necrosis.
Asafoetida (Hing)	Colophony (pine) resin, gypsum	Pure forms milky emulsion; burns cleanly	Colophony causes contact dermatitis & ulcers.

Ceylon Cinnamon vs. Cassia Bark

True Ceylon cinnamon consists of delicate, multilayered, paper-thin parchment quills with safe coumarin ($< 0.004\%$). Cassia consists of thick (1 – 3 mm), single woody hollow rolls containing **0.2 – 1.0% Coumarin**. Coumarin is epoxidized by CYP2A6 into **3,4-coumarin epoxide**, causing centrilobular liver necrosis and kidney injury.

6 Master Mnemonics for Fast Exam Recall

High-Yield Exam Mnemonics

- **D-R-O-P-S-Y (Argemone / Sanguinarine):** Dilation of capillaries (via NO) • Redness/rash • Oxidative phosphorylation uncoupled • Pitting lower limb edema • Sanguinarine inhibits Na^+/K^+ -ATPase • Yellow UV fluorescence on TLC.
- **F-U-N-D-S (Milk Rapid Screening Tests):** Formalin (Hehner's violet ring) • Urea (DMAB yellow-green) • Neutralizers (Rosolic acid rose-red) • Detergent (Methylene blue chloroform layer) • Starch (Iodine deep blue).
- **B-A-U-D-O-U-I-N (Vanaspati in Ghee):** By-product sesamol • Acid (HCl) hydrolysis • Unsaponifiable sesame oil • Dialdehyde: Furfural • Overlay turns **Rose-Red** • Indicates Vanaspati • Negative in pure ghee.
- **O-D-A-P (Khesari Dhal Neurotoxin):** Oxalyl radical (β -ODAP / BOAA) • Depolarization via AMPA • Arhar dhal substituted • Paraplegia (Neurolathyrism).

7 High-Yield Solved Question Bank

7.1 Multiple Choice Questions (MCQs)

- Which neurotoxin in Khesari Dhal causes neurolathyrism? (a) Sanguinarine (b) β -ODAP / BOAA (c) Coumarin (d) Alloxan [Ans: (b)]
- In Hehner's test for formalin in milk, the violet ring forms by reaction with which amino acid? (a) Methionine (b) Cysteine (c) Tryptophan (d) Phenylalanine [Ans: (c)]
- Baudouin test detects Vanaspati in ghee via condensation of furfural with: (a) Gossypol (b) Sesamol (c) Sanguinarine (d) Ricinoleic acid [Ans: (b)]
- Primary enzyme inhibited by Sanguinarine in Epidemic Dropsy is: (a) Acetylcholinesterase (b) Na^+/K^+ -ATPase (c) Cytochrome c oxidase [Ans: (b)]
- Melamine is added to milk to deceptively inflate: (a) RM value (b) Specific gravity (c) Crude protein in Kjeldahl test [Ans: (c)]
- Metanil yellow in pulses is confirmed when concentrated HCl gives: (a) Emerald green (b) Persistent magenta-pink (c) White turbidity (b)] [Ans: (b)]
- Holde's test detects mineral oil in vegetable oil based on: (a) Persistent milky turbidity after saponification (b) Acidity (c) Fluorescence [Ans: (a)]
- Toxic component in Cassia bark that induces centrilobular liver necrosis is: (a) Eugenol (b) Coumarin (c) Cinnamaldehyde (d) Safrole [Ans: (b)]
- Statutory FSSAI maximum limit of Total Polar Compounds (TPC) in frying oil is: (a) 10% (b) 15% (c) 25% (d) 40% [Ans: (c)]
- Papaya seeds are separated from genuine black pepper because they: (a) Sink in water (b) Float on water/alcohol (c) Turn blue with iodine [Ans: (b)]

7.2 Short Answer Questions (SAQs)

- **Q1. Explain the Baudouin Test principle.** Ans: Vanaspati contains mandated 5% sesame oil. Concentrated HCl hydrolyzes sesame sesamol into sesamol. Sesamol condenses with added furfural in acid medium to yield an intense, persistent **crimson-rose red** quinoid dye. Pure ghee gives no color.
- **Q2. What causes Epidemic Dropsy?** Ans: Consumption of mustard oil adulterated with *Argemone mexicana* oil containing **Sanguinarine**. Sanguinarine inhibits Na^+/K^+ -ATPase and triggers excess Nitric Oxide release, causing bilateral lower-limb pitting edema, secondary open-angle glaucoma, and cardiac failure.
- **Q3. Why does Kjeldahl test fail to detect Melamine?** Ans: Kjeldahl converts all nitrogen to $(\text{NH}_4)_2\text{SO}_4$ and multiplies by 6.38. Because melamine has 66.6% nitrogen, it deceptively inflates apparent protein without distinguishing peptide bonds from triazine rings.
- **Q4. How to differentiate Ceylon Cinnamon from Cassia?** Ans: Ceylon cinnamon has delicate multilayered parchment quills with safe coumarin (< 0.004%). Cassia has thick, woody single hollow rolls containing high **hepatotoxic Coumarin** (0.2 – 1.0%).
- **Q5. Role of Potassium Bromate in flour.** Ans: KBrO_3 oxidizes gluten cysteine –SH to disulfide crosslinks (–S – S–), trapping CO_2 and inflating bread volume. It is a banned IARC 2B renal carcinogen.

7.3 Solved Quantitative Numericals

Numerical 1: Kjeldahl Melamine Protein Correction

Problem: Milk powder has 0.7844% total nitrogen. (a) Find apparent protein ($F = 6.38$). (b) If 0.15% of powder weight is Melamine (66.63% N), calculate true protein.

Solution:

1. Apparent Crude Protein = $0.7844\% \times 6.38 = 5.004\%$.
2. Nitrogen contributed by 0.15 g melamine per 100 g = $0.15 \times 0.6663 = 0.0999\%$ N.
True Dairy Nitrogen = $0.7844\% - 0.0999\% = 0.6845\%$.
3. True Protein = $0.6845\% \times 6.38 = 4.367\%$. *(Melamine inflated protein by +14.6%)*.

Numerical 2: Reichert-Meissl (RM) Value Dilution

Problem: Suspected cow ghee has an RM value of 14.5. Pure ghee RM is 28.0 and Vanaspati RM is 0.6. Calculate % Vanaspati.

Solution:

$$\% \text{Vanaspati} = \frac{RM_{\text{pure}} - RM_{\text{sample}}}{RM_{\text{pure}} - RM_{\text{vanaspati}}} \times 100 = \frac{28.0 - 14.5}{28.0 - 0.6} \times 100 = \frac{13.5}{27.4} \times 100 = 49.27\%$$

8 Master Quick Revision Sheet & Pre-Exam Checklist

Food Matrix	Target Adulterant	Key Forensic Reagent & Diagnostic Observation
Grains (Wheat/Bajra)	Ergot sclerotia	Floats in 20% NaCl brine solution; clean sound grains sink.
Pulses (Arhar Dhal)	Khesari Dhal (β -ODAP)	Wedge-shaped trapezoidal seeds; boiling in water for 60 min detoxifies.
Pulses / Turmeric	Metanil Yellow	Conc. HCl $\rightarrow$ persistent magenta-red color ($\lambda \approx 530$ nm).
Pulses / Turmeric	Lead Chromate (PbCrO_4)	Ash + HNO_3 + KI $\rightarrow$ golden-yellow precipitate of $\text{PbI}_2 \downarrow$.
Flour (Maida)	Potassium Bromate	Spotting with 0.5% KI + 2M HCl $\rightarrow$ black-purple spots .
Milk	Added Water	Lactometer reading < 28 ; Freezing point shifts toward 0°C (norm: -0.54°C).
Milk	Starch / Flour	1% $\text{I}_2/\text{KI} \rightarrow$ intense deep blue-black inclusion complex.
Milk	Urea	DMAB reagent $\rightarrow$ intense bright yellow-green color.
Milk	Detergents	Methylene blue shake $\rightarrow$ lower chloroform layer turns blue .
Milk	Neutralizers	0.1% Rosalic acid in ethanol $\rightarrow$ rose-red color (pure is brownish-orange).
Milk	Formalin (37%)	Hehner's Test: Conc. $\text{H}_2\text{SO}_4 + \text{Fe}^{3+} \rightarrow$ violet-purple ring at interface.
Milk	Melamine (66.6% N)	HPLC-UV (240 nm); forms insoluble spherulites with cyanuric acid.
Ghee & Butter	Vanaspati	Baudouin Test: Furfural + HCl $\rightarrow$ crimson-rose red lower layer.
Mustard Oil	*Argemone* oil	Conc. $\text{HNO}_3 \rightarrow$ crimson ring; TLC gives orange UV fluorescence (366 nm).
Vegetable Oils	Mineral Oil	Holde's Test: Alcoholic KOH $\rightarrow$ persistent milky turbidity in water.
Vegetable Oils	Cottonseed Oil	Halphen Test: 1% Sulfur in CS_2 at $110^\circ\text{C} \rightarrow$ deep rose-red color .
Frying Oil (UCO)	High Polar Compounds	**FSSAI threshold: Total Polar Compounds (TPC) $\leq 25\%$.
Chili Powder	Sudan Dyes I-IV	Brick dust settles in water; lipophilic Sudan dyes detected by TLC.
Black Pepper	Dried Papaya Seeds	Floatation test: Papaya seeds float on water; peppercorns sink.
Cinnamon	Cassia Bark (Coumarin)	Single thick woody quill; high hepatotoxic coumarin (0.2 – 1.0%) .
Asafoetida (Hing)	Colophony Resin	Pure forms milky emulsion in water; burns with bright flame without black ash.

8.1 Pre-Exam Mastery Checklist

- ☐ **Khesari Dhal & β -ODAP:** AMPA receptor excitotoxicity, upper motor neuron lysis, spastic paraplegia.
- ☐ **Metanil Yellow:** Cleavage into toxic *p*-aminodiphenylamine, magenta-red test with conc. HCl.
- ☐ **Lead Chromate:** δ -ALAD inhibition, microcytic anemia, saturnism, yellow $\text{PbI}_2 \downarrow$ with KI.
- ☐ **Potassium Bromate & Alloxan:** Gluten $-\text{SH} \rightarrow -\text{S}-\text{S}-$, GLUT2 uptake, Fenton $\cdot\text{OH}$ diabetes.
- ☐ **Synthetic Milk:** Oil, detergent, urea, neutralizers, starch; rapid colorimetric detection tests.
- ☐ **Hehner's Test:** Formalin cross-links tryptophan in casein in presence of $\text{Fe}^{3+} \rightarrow$ violet ring.
- ☐ **Melamine Fraud:** 66.6% nitrogen, Kjeldahl flaw, [Melamine $\cdot$ Cyanuric acid] kidney stones.
- ☐ **Baudouin Test & RM Value:** Sesamolin $\rightarrow$ sesamol + furfural $\rightarrow$ crimson-red; RM drop to < 1.0 .
- ☐ **Argemone / Epidemic Dropsy:** Na^+/K^+ -ATPase inhibition, eNOS NO release, edema, glaucoma.
- ☐ **Holde's Test for Mineral Oil:** Saponifiable vegetable fats dissolve vs. non-saponifiable mineral oil turbidity.
- ☐ **Cassia vs. Ceylon Cinnamon:** Coumarin 3,4-epoxidation liver toxicity, single vs. multi-layered quills.
- ☐ **Quantitative Calculations:** Melamine nitrogen correction, RM value dilution, lactometer water addition.